NOVEX HACKATHON 2026

TITLE PAGE

PROBLEM STATEMENT TITLE —

PCP Care: Unified Pathology & Sample Tracking Platform

PROBLEM STATEMENT —

- Booking tests, sample pickup, and report delivery happen across disconnected channels.
- Patients and labs have no live tracking for blood and urine samples during transit.
- Lab reports use difficult medical terms that patients cannot easily understand.

PS CATEGORY —

HealthTech / Healthcare Logistics

TEAM NAME —

Erro404



PCP Care Platform

One Platform. Complete Pathology Care.

5 Portals	100%	AI Helper
All Roles Connected	Barcode Tracked	Simple Explanations

IDEA TITLE: PCP CARE PLATFORM

Proposed Solution (Describe your Idea/Solution/Prototype)

1. Detailed Explanation of Proposed Solution

- **Single Platform:** Connects patients, sample collectors, certified labs, and doctors in one web app.
- **Patient Portal:** Search tests, compare prices across local labs, and book home collection.
- **Collector App:** Shows pickup requests, optimal travel routes, and barcode scanning tools.
- **Lab Dashboard:** Verifies incoming sample tubes, manages tests, and uploads PDF reports.
- **Doctor Access:** Doctors can view patient test history directly with patient approval.

2. How It Addresses the Problem

- **No Lost Samples:** Barcode scanning at patient doorstep and lab intake ensures 100% custody tracking.
- **No More Waiting:** Fast digital handoffs cut total testing time by almost half.
- **Easy to Read:** AI turns confusing clinical values into simple color-coded cards (Normal / High / Low).

3. Innovation and Uniqueness

- **AI Report Explainer:** Explains what tests mean in simple words without diagnosing or prescribing.
- **Secure Doctor Sharing:** Patient generates a 4-digit temporary code to give doctors instant review access.
- **Fair for Local Labs:** Open marketplace with small 8–10% fee instead of 35%+ aggregator cuts.

HOW IT WORKS (5 SIMPLE STEPS)

Step 1	**Book Test:** Patient chooses lab, test package, and home slot.
Step 2	**Collect & Scan:** Phlebotomist collects sample and scans barcode tube.
Step 3	**Lab Intake:** Lab checks sample barcode, runs test, uploads PDF.
Step 4	**AI Explains:** System extracts numbers and explains values in plain words.
Step 5	**Doctor Consult:** Patient shares secure link with doctor for fast consultation.

⚠️ **Important Safety Rule:** The AI only helps users understand medical terms and test ranges. It does not give medical advice, write prescriptions, or replace a licensed doctor.

TECHNICAL APPROACH

Technologies to be Used & Implementation Methodology

1. TECHNOLOGIES TO BE USED

- **Frontend:**
  HTML5, Tailwind CSS, and Thymeleaf. Works smoothly on both mobile phones and desktop browsers.

- **Backend & APIs:**
  Java 21 with Spring Boot 3. Fast, stable, and handles multiple user roles securely with Spring Security.

- **AI Integration:**
  Spring AI connected to OpenRouter. Reads PDF lab values and explains them safely using strict reference limits.

- **Database & File Storage:**
  MySQL for booking and user records. Cloudinary for storing encrypted PDF reports with expiring access links.

- **Security & Auth:**
  JWT token login, role-based access for 5 user types, and encrypted communication over HTTPS.

Build & Run: Managed with Apache Maven and ready to run inside Docker containers.

2. SYSTEM ARCHITECTURE & FLOW

Tier 1: User Interfaces (Web & Mobile)
Dedicated portals for Patients, Collectors, Labs, and Doctors.

Tier 2: Security & Login Layer
Spring Security checks user permissions and manages token-based login.

Tier 3: Core Application Services
Booking engine, collector route dispatch, and lab report upload handlers.

Tier 4: Spring AI Report Processor
Extracts test numbers, compares with normal ranges, creates simple summaries.

Tier 5: Database & Document Storage
Reliable MySQL data storage and secure cloud PDF vault.

Working Prototype: All modules are already connected and tested with simulated test bookings and real PDF report parsing.

FEASIBILITY AND VIABILITY

Analysis of Feasibility, Challenges & Practical Strategies

1. ANALYSIS OF THE FEASIBILITY OF THE IDEA

⚡ Technical Feasibility

- Built on standard Java Spring Boot and MySQL.
- Loads fast (<1 second) on regular home internet.
- Works on any smartphone or laptop without installation.

🛵 Operational Feasibility

- Collectors only need a phone camera to scan barcodes.
- Labs can start using the portal in 15 minutes.
- No costly custom machinery or hardware required.

📈 Financial Viability

- Charges labs only 8–10% per test (vs 35%+ legacy apps).
- Saves money for patients while keeping local labs profitable.
- Sustainable and scalable without heavy cash burn.

2 & 3. POTENTIAL CHALLENGES, RISKS & OVERCOMING STRATEGIES

Potential Challenge / Risk	Area	Strategy to Overcome Challenge
AI Making Medical Mistakes Giving wrong medical advice or diagnosing a condition.	Clinical Safety	**Strict Rules:** The AI only checks if numbers fall inside official lab reference tables. It never suggests treatments or diagnoses, and shows a mandatory doctor consultation prompt.
Sample Damage or Delay Blood sample getting delayed in traffic or spoiled in heat.	Logistics	**Barcode Tracking & 45-Min Limit:** Every handover is scanned with timestamp. The app groups pickups by nearby areas so samples reach the lab within 45 minutes inside cold bags.
Patient Data Privacy Confidential diagnostic reports getting seen by strangers.	Security	**Encrypted & PIN Protected:** All reports are encrypted in storage. Only the patient and their approved doctor can view reports via temporary secure access links.

IMPACT AND BENEFITS

Impact on Target Audience & Practical Real-World Benefits

1. IMPACT ON TARGET AUDIENCE

- **Patients & Families:**
  Understand test results in simple words without stress; book tests easily from home; save 20–30% on test costs.

- **Local & Neighborhood Labs:**
  Get more customers online and manage sample pickups without paying huge commissions to large aggregators.

- **Doctors:**
  View clear, organized past test records online instead of deciphering unclear paper photos sent over chat.

2. BENEFITS OF THE SOLUTION

🌍 Social Benefits

- Helps common people understand blood sugar, cholesterol, and kidney markers without panic.
- Promotes earlier doctor visits before health conditions become serious.

💰 Economic Benefits

- Protects local certified laboratories from being forced out by large corporate chains.
- Lowers healthcare spending for everyday families through transparent competition.

🌱 Environmental Benefits

- Grouped pickup routes cut unnecessary motorbike trips across town, saving fuel.
- Completely digital workflow removes thousands of paper reports and plastic envelopes every month.

-40%	0%	100%	25%+
Faster Results	Lost Samples	Digitally Tracked	Cost Savings

Practical, low-cost solution with high impact on daily healthcare access.

RESEARCH AND REFERENCES

Details / Links of the Reference and Research Work

REFERENCE WORK & FOUNDATIONS

1. Healthcare & Diagnostic Studies

• **WHO Clinical Diagnostics Report:** Highlights that over 70% of clinical decisions depend on lab tests.
• **Journal of Clinical Pathology:** Confirms that over 60% of diagnostic errors occur during sample pickup, transit delays, and label mix-ups.

2. Laboratory & Transport Standards

• **ISO 15189 Standards:** Guidelines for medical laboratory quality, digital traceability, and specimen integrity.
• **NABL Guidelines:** Protocols for safe sample collection, cold-chain transit, and standardized test result reporting.

3. Digital Health & Privacy Guidelines

• **Ayushman Bharat Digital Mission (ABDM):** Guidelines on digital health records, patient consent, and secure data exchange.
• **HIPAA & DISHA Principles:** Best practices for protecting sensitive health records with role-based permissions and encryption.

4. Technology & Safe AI Implementation

• **Spring AI Framework:** Official docs for building secure Java AI systems with strict output schemas.
• **Lancet Digital Health (2024):** Research confirming that simple plain-language explanations help patients follow doctor instructions better.

TEAM ERRO404 · NOVEX 2026 ROUND 2 PPT SUBMISSION

Concise, bullet-point format following the official 6-slide template.

✓ READY TO SUBMIT